



Public

Making Data Centers Sustainable: Reducing the Environmental Impact of SAP Data Centers

Owners: *DLS Cloud Infrastructure and Environmental Sustainability (CIES) & Global Sustainable Operations & Energy (GSO&E)*

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1. Creating Positive Impact: SAP's role

At SAP, our mission is to help the world run better and improve people's lives by embedding sustainability into our business and the businesses we serve.

Our strategic approach to sustainability focuses on three pillars: climate action, circular economy, and social responsibility. We are committed to achieving net-zero emissions along our value chain by 2030, in line with a science-based 1.5 °C future.

Additionally, we aim to operate free of single-use plastics, strive toward zero e-waste in our operations, and respect and advance human rights across our value chain.

To achieve this, we adopt two complementary frameworks:

1. **SAP as an enabler:** Providing products and services that support our customers' work to meet their sustainability challenges and allow them to identify new opportunities.
2. **SAP as an exemplar:** Leading by example in our own sustainable business operations and practices.

This approach allows us to learn first-hand and therefore guide our customers more effectively on their own sustainability journeys.

2. SAP's Data Centers: A Program of Continuous Improvement

SAP is committed to sustainability across our entire business. At the heart of this commitment are SAP Cloud services — the infrastructure that powers everything we do. We pay particular attention to our data centers, where we have the greatest opportunity to reduce our environmental impact.

Data centers pose significant environmental challenges due to their high energy and water consumption, particularly for cooling purposes. Data centers account for [3% of global electricity use](#), with projections suggesting this could rise to [10% by 2030](#). The increasing demands from AI workloads further intensify cooling and water needs. Moreover, data center infrastructure contributes to carbon emissions, resource depletion from hardware production, and e-waste.

With the rapid acceleration of cloud adoption, SAP's data centers, both owned and co-located (where our servers are housed in the same data center as other companies' servers), have emerged as a central focus of our sustainability initiatives. We take a holistic approach to sustainable data center management, from efficient sourcing and operation to responsible decommissioning hardware ensuring alignment with our broader sustainability objectives.

This document will guide you through SAP's efforts to enhance environmental sustainability within our data centers, while maintaining operational excellence.

2.1 Minimizing Environmental Impact Through Strategic Location Selection and Optimizing Workload Deployment

SAP begins by identifying the best location to host our and our customers' workloads. We intentionally select data centers for new workload deployment in regions that maximize energy efficiency and minimize reliance on fossil fuels. Consideration of these elements allows us to support local renewable energy markets and encourage the development of sustainable infrastructure. This involves analyzing and mapping regional energy profiles to identify optimal locations. In addition to sustainability considerations, we review criteria that balance technical requirements and commercial viability.

2.2 Selecting Energy Efficient Hardware

SAP promotes the use of IT hardware that uses less energy and therefore generates lower emissions. Our servers, for example, use Titanium class Power Supply Units (PSUs)—which have the highest power efficiency rating (90%+ efficiency at converting electricity into usable power compared to bronze class which has an 80% efficiency).

We also select the Central Processing Units (CPUs) that meet our performance requirements with the lowest Thermal Design Power (TDP) rating to achieve the lowest energy consumption.

Finally, SAP selects server racks with more spacing to allow better airflow and therefore more effective cooling of the server. To ensure the procurement of the most efficient hardware, SAP embeds energy efficiency requirements into contracts with all our server suppliers.

2.3 Optimizing Data Center Cooling

When IT appliances are used, they generate heat, which can cause malfunctions if the ambient temperature surpasses their recommended operating range. To prevent hardware device failure due to excessive heat, consistent cooling is essential.

Several measures can reduce the carbon footprint associated with cooling in data centers, including:

1. Minimizing the amount of cooling required by operating at higher temperatures.
2. Designing a data center architecture that optimizes the existing cooling capacity.

The specific cooling requirements known as cold loads in our data centers largely depend on the respective rack power density. To address these varying and increasing demands, dedicated retrofit initiatives are being implemented across our existing facilities.

Less Cooling

Reducing data center cooling requirements leads to lower electricity consumption, not only cutting costs but also minimizing environmental impacts.

To optimize energy use, SAP has raised the maximum temperature of the air entering the server to 27°C in most of its data centers in line with the American Society of Heating, Refrigerating, and Air-Conditioning Engineers "recommended range".

Furthermore, SAP is exploring partnerships with manufacturers to support further increases in intake temperatures, aiming to enhance efficiency and uphold sustainable practices in data center operations.

To efficiently manage the rising thermal loads of future hardware generations, SAP has also analyzed potential Direct Liquid Cooling (DLC) opportunities to identify optimal technology. The specific aim of this initiative is to retrofit our SAP-owned data centers into a hybrid cooling system that combines traditional air cooling with advanced liquid cooling.

Increasing Cooling Efficiency

Improving cooling efficiency in data centers is vital for cost reduction, carbon footprint minimization, equipment reliability, and high-density server deployment. SAP has adopted various measures to enhance data center cooling efficiency, including managing airflow using a hot and cold aisle layout for server racks. A containment system isolates the hot aisles and cold aisles from each other, preventing hot and cold air from mixing. This arrangement generates less heat in the hot aisles, whilst cold aisles temperatures remain low, thus putting less strain on the air conditioning system resulting in cost savings, environmental benefits, and improved performance.

In addition to implementing measures to increase cooling efficiency in data centers, SAP employs the following strategies to effectively control energy loss:

- **Thermography Process:** SAP uses infrared cameras to detect hot and cold spots in data centers, indicating the need for increased airflow or improved cabinet sealing to enhance energy efficiency.
- **FLIR Tool:** The Forward-Looking Infrared tool detects inefficient airflows, which helps prevent hardware failures caused by extreme temperatures, minimizing customer downtime. It enables SAP to produce annual or incident-related reports on data centers, both owned and co-located. These reports are analyzed for corrective actions.
- **Temperature Monitoring and Management:** SAP is expanding the installation of temperature sensors on the front and back sides of each rack in all data halls for continuous monitoring. This setup provides detailed comparisons between desired and actual temperatures, mapping the results to identify hot spots and analyzing temperature trends, thereby facilitating corrective and preventive actions.

By actively deploying these thermography and continuous monitoring processes, SAP is actively identifying concrete opportunities to reduce the required cooling energy across our live environments, ensuring that our daily operations reflect our sustainability ambitions.

2.4 Leveraging Renewable Energy to Reduce GHG Emissions

Since 2014, all SAP data centers and data center regions, whether owned or co-located, have been powered by 100% renewable electricity, significantly reducing greenhouse gas emissions. Building on this, SAP is actively shifting from Energy Attribute Certificates (EACs) to mid- and long-term Power Purchase Agreements (PPAs), with markets like Germany expected to cover up to 90% of power consumption through PPAs by November 2026. These efforts are complemented by on-site renewable energy generation, including the installation of solar panels.

A key part of SAP's decarbonization journey is the transformation of its Walldorf and St. Leon-Rot headquarters, where fossil-based natural gas is being phased out. Previously used in combined heat and power systems that also supported DC cooling, this approach is being replaced with more sustainable solutions such as thermal storage (including ice storage) and battery energy storage systems, enabling more efficient and flexible energy use.

2.5 Balancing Water Use and Energy Efficiency

Balancing PUE (Power Usage Effectiveness) and WUE (Water Usage Effectiveness) in data centers is challenging because improving one can sometimes negatively impact the other.

For example, cooling systems are a major energy consumer in data centers and can greatly affect PUE. Using efficient air cooling might improve PUE by reducing energy use but often requires more water, which worsens WUE. Conversely, opting for cooling methods that use less water, like air or evaporative cooling, can increase energy consumption, potentially leading to a higher PUE.

Moreover, external factors such as climate and location influence the ability to efficiently balance PUE and WUE. In hot climates, more water might be needed to keep systems cool, while in water-scarce regions, using energy-intensive cooling might be the only option.

Finding a balance involves technology choices, geographical considerations, and constant innovation. It requires data centers to strategically manage resources, seeking new technologies that can enhance both energy and water efficiency simultaneously, ensuring sustainability.

Adiabatic Cooling

To minimize the carbon footprint of our data centers, SAP predominantly employs adiabatic (evaporation) cooling systems. Using evaporation to cool the ambient air significantly reduces the load on electrical compressors. The challenge is achieving high energy efficiency while optimizing the consumption of water.

Optimization Through Closed Loops

To address this tension, SAP is pursuing an advanced water management strategy, using closed water circuits to cool server racks to minimize water use.

Performance Against Industry Standards

Our consumption data reflects the success of this balanced approach. While traditional water-cooled data centers can consume up to 26 million liters of water annually per megawatt (MW) of cooling capacity, SAP has optimized its operations to consume only between 6 and 11 million liters per MW per year. This varies according to the specific technologies deployed but proves that energy efficiency does not have to come at the expense of excessive water usage.

2.6 Re-Using Waste Heat from Data Centers

When utilizing this waste heat, the data center must not be viewed as an isolated unit. SAP pursues a holistic approach where the data center acts as an active 'prosumer' - serving as a central energy provider deeply integrated into the sector coupling of the entire headquarters campus.

Thermal energy is a significant byproduct of data center operations. Traditional systems emit this heat into the atmosphere, but SAP recognizes it as a valuable resource that can be reintegrated to reduce energy consumption in other systems. Harvesting this energy requires us to transform the cooling infrastructure.

Transitioning from Air-Air to Air-Water Cooling

To make waste heat usable, it must be captured in a medium that can transport energy efficiently. SAP is currently executing comprehensive retrofit measures to convert legacy cooling systems. Since water has a significantly higher heat capacity than air, we are transitioning from pure air-air cooling to air-water cooling systems allowing the thermal load from the servers to be transferred to a water circuit. This heated water can then be fed into heating networks, reducing the need for buildings to generate their own heat, saving money, and reducing fuel consumption which decreases emissions.

2.7 Circularity and Hardware End of Life Strategies

SAP is working toward zero e-waste in its operations by [2030](#), by ensuring responsible end-of-life management of equipment. The company consistently diverts over [90% of e-waste](#) from landfills and incineration, primarily by sending decommissioned data center devices to certified IT Asset Disposition (ITAD) providers, as they are rarely used internally. Through these partners, valuable materials like rare earth metals are recovered to reduce the need for new extraction and lower environmental impact. SAP tracks progress through detailed device-level disposal reports and supports these efforts through close collaboration across teams such as the Global Procurement Office, Corporate Sustainability, and key suppliers.

2.8 Strategic Engagement with Hyperscalers

Hyperscalers are large cloud computing providers such as Amazon Web Services, Microsoft Azure, and Google Cloud. They represent an important part of SAP's supply chain. The emissions they generate from the services we purchase are part of our carbon footprint. We therefore collaborate closely with them on environmental sustainability topics. Our strategic engagement is two-fold:

- At a corporate level we align sustainability agendas such as net-zero targets, and best practice. This is straightforward where their net-zero ambitions align with our own 2030 target (Google and Microsoft), but is more challenging in the case of Amazon, which is targeting 2040. To overcome the mismatch in ambition and potential differences in priorities and the speed of change, SAP is pursuing strategic cooperation at Chief Sustainability Officer (CSO) level to define mitigation measures.
- At a technical level, SAP works with hyperscalers to gain as much visibility into the data centers they run as possible, working towards the automatic collection of detailed energy use and emissions data. This allows SAP to measure and fulfill reporting and emissions reduction commitments.

SAP is also onboarding new hyperscalers including IBM and StackIT, aligning with them on the strategic and technical levels outlined above.

3. Compliance and Alignment with Global Regulatory Bodies

All SAP owned data centers have achieved [ISO14001-certification](#). This certification ensures that SAP systematically manages its data center operations and environmental responsibilities, monitors environmental performance, and strives for continuous improvement in its environmental impact. Additionally, to maximize energy efficiency and continuously enhance energy performance, SAP has implemented an ISO 50001-certified [Energy Management System](#) at these data centers. This certification ensures that they operate with optimal energy efficiency, further aligning with SAP's sustainability goals.

We align our sustainability practices with global and national standards by reporting on the [EU Energy Efficiency Directive \(EED\)](#), the [German Energy Efficiency Act \(EnEfG\)](#), the [EU Taxonomy](#), the [EU Code of Conduct for Energy Efficiency in Data Centers](#). SAP's data center team is simultaneously working with regulating bodies like the European Union to set the future standards of green information technology. To further support it, SAP is also a member of the [Climate-Neutral Data Center Pact](#) community.